

JOHANNESBURG INTL, South Africa

WMO: 683680

Lat: 26.143S

Lon: 28.235E

Elev: 5561

StdP: 11.97

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	35.9	6.9	9.6	57.6	10.5	11.5	56.1	23.2	49.3	21.1	50.3	9.6	210	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	84.4	58.6	82.5	58.8	80.6	59.2	66.9	74.9	65.6	73.9	64.6	73.3	11.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	111.8	70.2	62.9	105.8	68.6	62.3	103.4	68.0	35.0	75.5	33.9	74.2	33.0	73.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.8	16.8	DB	27.5	88.1	3.8	2.2	24.8	89.6	22.5	90.9	20.4	92.1	17.6	93.7	(4)
(5)			WB	24.0	68.7	3.1	2.1	21.8	70.2	20.0	71.4	18.2	72.6	16.0	74.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	67.7	67.8	65.8	61.0	56.2	51.8	50.9	56.1	61.8	64.6	65.7	67.3	(6)
(7)		DBStd	7.71	3.74	3.55	3.83	4.41	5.28	4.62	5.06	5.99	6.46	5.77	5.11	4.02	(7)
(8)		HDD50	116	0	0	0	1	12	32	50	17	4	0	0	0	(8)
(9)		HDD65	1869	16	12	36	133	274	396	437	281	134	78	50	22	(9)
(10)		CDD50	4264	550	500	489	331	206	86	78	205	357	453	473	536	(10)
(11)		CDD65	541	102	92	60	12	2	0	0	4	38	65	73	93	(11)
(12)		CDH74	3689	550	517	350	69	10	0	0	28	352	641	584	588	(12)
(13)		CDH80	514	71	72	38	1	0	0	0	37	119	96	80	80	(13)
(14)	Wind	WSAvg	8.6	8.3	7.7	7.5	7.5	7.4	7.8	7.8	9.1	10.0	10.4	10.0	9.5	(14)
(15)	Precipitation	PrecAvg	28.6	4.8	3.8	3.7	1.9	0.6	0.3	0.1	0.3	0.9	3.0	4.3	4.3	(15)
(16)		PrecMax	42.9	13.3	10.4	13.1	5.1	3.9	2.4	0.9	1.8	6.9	7.8	9.1	8.7	(16)
(17)		PrecMin	17.6	1.7	0.5	0.6	0.1	0.0	0.0	0.0	0.0	0.0	0.5	0.9	1.2	(17)
(18)		PrecStd	5.7	2.4	2.1	2.2	1.2	0.9	0.6	0.2	0.4	1.2	1.8	1.9	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.7	85.6	84.0	78.9	75.3	69.8	70.0	77.1	83.9	86.3	86.2	85.8	(19)
(20)			MCWB	61.5	59.7	58.5	57.2	53.3	46.8	46.3	51.8	53.7	56.1	58.7	59.8	(20)
(21)		2%	DB	82.2	82.2	80.5	75.6	71.7	66.5	67.0	73.7	80.7	83.4	82.8	82.4	(21)
(22)			MCWB	61.8	60.9	60.1	56.0	51.0	46.7	45.8	50.3	53.0	55.6	59.2	60.6	(22)
(23)		5%	DB	79.2	79.3	77.6	73.5	69.4	64.5	64.6	71.3	78.5	80.6	80.3	80.0	(23)
(24)			MCWB	62.6	61.7	59.8	55.8	50.9	46.1	45.5	49.4	52.6	56.4	59.2	61.4	(24)
(25)		10%	DB	77.1	77.1	75.3	71.2	66.9	62.5	62.3	68.4	75.4	77.5	77.3	77.3	(25)
(26)			MCWB	62.7	61.9	59.9	55.6	49.9	45.8	45.0	48.5	51.9	56.2	59.0	61.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	67.4	67.4	62.7	58.3	52.2	51.8	56.6	59.4	63.5	65.7	68.4	(27)
(28)			MCDWB	77.1	76.0	72.3	70.2	66.6	61.4	63.6	67.6	73.2	74.5	75.8	76.8	(28)
(29)		2%	WB	66.7	65.8	65.2	60.8	55.6	50.3	49.5	54.2	57.4	61.7	64.2	66.4	(29)
(30)			MCDWB	74.8	74.3	72.5	68.5	64.6	60.1	59.8	67.4	72.1	73.3	74.7	74.6	(30)
(31)		5%	WB	65.4	64.7	63.7	59.4	54.0	49.1	48.0	52.5	55.9	60.2	62.8	64.9	(31)
(32)			MCDWB	73.6	73.4	71.4	67.0	63.6	59.0	58.9	65.6	70.3	72.2	73.2	73.4	(32)
(33)		10%	WB	64.4	63.8	62.3	58.0	52.5	47.8	46.5	50.8	54.5	58.9	61.7	63.6	(33)
(34)			MCDWB	72.9	72.1	69.9	66.0	62.3	58.0	58.2	63.2	68.8	70.4	71.8	72.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.5	17.9	17.9	18.6	20.1	20.9	21.6	22.5	23.2	22.0	19.9	18.4	(35)
(36)			MCDBR	21.1	21.6	21.4	21.3	20.9	22.2	22.9	23.6	24.2	24.7	23.5	21.8	(36)
(37)			MCWBR	8.5	8.5	8.3	8.9	9.2	10.5	10.5	10.4	9.7	9.6	8.9	8.5	(37)
(38)		5% WB	MCDBR	17.9	17.7	17.4	17.6	18.5	19.0	20.6	22.0	23.1	22.1	20.3	18.5	(38)
(39)			MCWBR	8.5	8.2	8.5	8.9	9.6	10.2	10.8	11.0	10.5	10.6	9.7	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.383	0.385	0.373	0.326	0.290	0.273	0.274	0.326	0.366	0.400	0.362	0.370	(40)	
(41)		taud	2.324	2.320	2.334	2.440	2.519	2.542	2.525	2.370	2.271	2.212	2.356	2.354	(41)	
(42)		Ebn at Noon	305	299	292	293	292	291	295	288	290	291	309	309	(42)	
(43)		Edh at Noon	44	43	40	33	27	25	27	35	42	47	42	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2100	1985	1759	1507	1377	1250	1358	1599	1905	2041	2132	2152	(44)	
(45)		RadStd	223	177	160	133	72	60	36	74	89	141	191	128	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157	
(47)	Regional (12 neighbors)		+1.19	+1.35	N/A	N/A	N/A	N/A	-326	+421	+157	(47)

Nomenclature: See separate page